

Brijodison

Engineering Notebook Element I

Engineering Service Learning

Data and Analysis

Label	Aesthetics (out of 10)	Handle Comfort (out of 10)	Time to Assemble (seconds)	Maximum Angle Bent (degrees)
Colton King	3	9	8.92	<5
Mr. Kiser	3	7	20.97	10-15
Connor Harris	7	8.5	21.51	<5
Cole Hawkins	3	3	29.99	<5
Nick Blucher	7	7	8.08	5-10
Average	4.6	6.9	17.894	<5 (mode)

On average, we did reduce the aesthetics of the cart, but this would be relatively easily fixable simply by painting the product or using the production materials. Robotics student and primary stakeholder, Colton King noted that the cart would be far more aesthetically pleasing if it were painted a solid black, fitting with the cart. We did not test any modifications to the aesthetics because they were far less important on our criteria list and they would potentially impede the rapid development of the product (e.g. glue may not stick to the PVC if we painted it and sanding the paint would likely leave scuffs that would be counterintuitive).

In terms of the handle's comfort, the product did very well, scoring a 6.9 out of ten on average. We would consider this to be a fair success, but we would attempt to revise the handle in the future while finishing off the product. In terms of qualitative feedback, Cole Hawkins noted that the handle of the cart was firm. He considered this to be an ergonomic benefit of the original handle. Other qualitative feedback showed the exact opposite. Nick Blucher noticed that the vertical handle maintained a solid balance between ergonomics (firm) and comfort (soft). We could consider making handle covers for a final product, allowing the user to switch between various handles depending on their preference.

The average time the users took to completely assemble and reassemble the product was 17.894 seconds (8.947 seconds on average to do one). This is an absolute success for us since Mr. Kiser noted in an interview that he would like the product to be able to be removed in under ten seconds, and this was possible on average. We completely succeeded in this aspect—although we could allow it to be assembled quicker, this revision takes least priority because it isn't immediately required.

The mode of the maximum angle of the participant's back is less than 5 degrees, but we noticed a few outliers that have some significance. Participants who were noticeably taller than the average were more likely to have to bend over to precisely put the clamp onto the lip of the base. As a result, certain primary stakeholders (e.g. Mr. Kiser) may have difficulty using this product the first few times because they would have to bend far more than the average while assembling the product. We should revise the process to involve more thorough testing to better measure this data (taking some sort of data over time would be very useful to allow the user to adapt their technique).

We received data from the force plate that indicated it would take a minimum of 10 Newtons to start the cart moving, but we would ideally like to do more tests in the future that would vary the load on the cart and give us more accurate data as to the effectivity of the cart on a variety of loads. As of now, a 10 Newton force to start the cart moving with its typical load seems very reasonable, and our qualitative feedback from Coach Sanford indicates that the cart was particularly easy to navigate forward and backwards.

We also used the force plate to measure the force required to break the grip of the product as indicated in the previous element. We discovered that it took approximately 55 Newtons to break the grip after substantial revision to prevent it from sliding. During our initial revision of the product, where nothing held the product to the cart, it took a near infinitesimally small force to accelerate the product (measured approximately 3 Newtons, but the force plate does not seem to be particularly apt for such problems that require precise measurements with low forces). The coefficient of static friction between the cart and the PVC was originally very small because both are relatively smooth, so friction did very little to hold the two together especially at a mechanical advantage with such a large lever arm. As indicated by the feedback from Jared Lawson, a graduate student at Vanderbilt with a B.S. in Mechanical Engineering, a solution was absolutely essential because the stability problem was evident by qualitative analysis.

We disregarded the weight of the product because the final product will be built out of a far denser material, so we didn't think the prototype would be very useful for this. We found the dimensions to be 1.21 meters by 0.11 meters by 0.03 meters, estimating a volume of 0.003993 m^3 , which would produce a product of mass 31.34505 kg as estimated using a density of 7850 kg/m^3 . This is fairly inaccurate because the volume is the volume of the smallest box the prototype could fit in, not the prototype itself. We should construct another prototype to better

measure this quality, but at the very least, this data tells us we would have to use some weight saving methods when producing the real prototype. Overall, the testing appeared to indicate that the prototype was a success, and our qualitative data from field expert Peter Hermann indicates the same, finding its simplicity reduces sources of error and represents a strong foundation.